Single Oral Practice 2

Module 5 - Single Oral Administration · Daily practice sheet · No date

Name _____ Date _____

A 10-mg single dose of a drug is given orally to a patient. The drug is about 80% absorbed and the equation that fits the observed data is $C_p = 70$ ng/mL $(e^{-0.1t} - e^{-0.3t})$. [Rate constants are in hr^{-1}]. Determine the following:

Part 1 Volume of distribution

Part 2 Total body clearance

Part 3 Half-life of elimination

Part 4 Half-life of absorption

Part 5 Maximum plasma concentration

Part 6 Maximum plasma concentration if the dose were increased to 20 mg

Part 7 Half-life of elimination if the clearance were reduced to 8.57 L/hr, assuming no change in apparent volume of distribution

Part 8 Maximum plasma concentration for a 10 mg dose if the clearance were reduced to 8.57 L/hr, assuming no change in apparent volume of distribution